Computer Graphics Project #3
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How to Play

The ship starts with 0 speed. This speed can be gained by pressing W. Holding shift increases speed gained by pressing W. Speed maxes out at 120. Speed decreases when W is not held. S can slow down the ship, or reverse it if held long enough. Negative speed also balances out back towards 0 eventually. The cursor must be clicked into the tab in order for the camera to be moved. Asteroids will damage the ship, while stars will heal it. Being hit by an asteroid after your health bar reaches 0 will stop the program from rendering, meaning you have lost.

- **Dillon Fiske**
  - **Asteroid Collision Handling)**
    - I implemented asteroid collision detection creating a bounding box in front of the camera that I used to represent a ship and I created hitboxes for each of the asteroids. When an asteroid bounding box collides with the ship's bounding box, then the position of the colliding asteroid is reset to a random position at negative five-hundred on the z-axis. This allows for the asteroid to re enter the scene from a distance. Additionally, Diana added a screen shaking function that temporarily moves the camera up and down each time a collision occurs.
  - **Infinite Asteroids)**
    - I achieved the goal of making the asteroids seem infinite by creating a function that uses a for loop to create many asteroid meshes, each with a unique drift pattern. The drift pattern of an asteroid can be either a sinusoidal, linear, parabolic, or irregular function, each with randomized parameters generated by Math.random(). The drift pattern of an asteroid is decided by the asteroid's creation order using i%4, where i is the loop index. Moreover, every asteroid moves along the

z-axis until either its z position is greater than or equal to six hundred or it collides with the ship, then its position is reset to a random position at the start. This gives the asteroid field a kind of infinite feel.

- **Asteroid Texturing)**
    - To go with the space theme, I used a texture loader to load in textures for the asteroids. First, I created the THREE.JS texture loader to be used to load in the image of the asteroid. Next, I downloaded a stock image of an asteroid from the internet and moved it into the repository. Finally, inside of the for loop used for the creation of the asteroids I set the texture of each asteroid to the asteroid image using the texture loader.
- **Space Texturing)**
    - To make the model look more like outer space, I used a texture loader to load in the background. First, I created the THREE.JS texture loader to be used to load in the image. Next, I downloaded a stock image of space from the internet and moved it into the repository. Finally, I set the background of the scene to the space image using the texture loader.

- **Matthew Thomas**
    - **Github)**
        - I set up the Github repository as well as created the Kanban board and put some issues on it, which we then assigned to ourselves.
    - **Star and Asteroid Shapes)**
        - I used a shaderMaterial on every asteroid and star that randomizes the scale of the points on each object. This created a somewhat random shape for every object in space, as it gave them jagged indentations and protrusions.
        - Stars have an “aura” around them, which is a low opacity copy of the star that attaches to it to give it the appearance of a glow.
    - **Shader with Textures)**
        - I integrated Dillon’s texture importing into the shaderMaterial and provided a texture for the stars to use, so that stars and

asteroids both can have random shapes and also have textures via the shaderMaterial.

- **Infinite Scrolling Objects**)
    - I built on Dillon's infinite asteroids by making it so that rather than a tunnel of asteroids and stars scrolling by, it instead randomly places asteroids and stars in random positions within a sphere around the starting point.
    - To make stars and asteroids seem infinite, it then checks a large radius around the ship. Any objects outside of that radius are randomly repositioned to a point in front of the ship, within the camera view. This achieved the effect of "new" stars and asteroids being rendered in as the ship travels forwards, and because it is a sphere around the ship, there will always be objects no matter what direction you go in.

- **Diana Livengood**
    - **Camera and ship movement**
        - I tried several approaches to the camera movement, then realized I was being stupid and that you wanted this to be first person. As such I decided to just take my working camera controls and apply them to the ship, then copy the ship's movement. Though now I'm not sure why the ship is still here. The rear view camera's movement was much easier to add once I had the foundations down, as I just had to copy it to the camera and then turn it 180 degrees. Implementing delta time into the movement was really tricky, as I wasn't 100% sure on when and where to add it. Additionally it messed up the pre-established values I had, which took lots of re-tuning. In the future I will begin with delta time instead of adding it after.
        - Camera movement I implemented by calculating the change in X/Y values of mousemovement instead of using an imported library. I'm much happier with this implementation but it still needs some work.
    - **Health Bar (Above and beyond - Line 493)**
        - The health bar is just two graphics objects placed over each other in the group. When the ship takes damage or is healed the "fill" bar has its X scale changed, and is moved within the

group to account for the change. I added the health bar to the camera so that it follows the camera around everywhere.

- **Rear View Camera**
    - Two perspective cameras are created. The rear view camera has its perspective cut out into a small square, and uses a view port to restrict its display to that corner. The scissor test is used to prevent it from overwriting the main camera.
- **Collision**
    - Just uses bounding boxes to detect collisions. There's definitely optimizations that could be made here (checking collision only on player movement, removing destroyed asteroids/stars from the world space AND the asteroid/star list) that I don't feel confident implementing as the generation for those objects is not my code. A sound effect plays during collision.